

Friday-Jan-24-2025

Hints about HW#1, (1) In P & 13 Eqⁿ for $h \parallel km$,
R is R_u (universal gas constant) $R_u = 8.314 \left(\frac{J}{kmol} \right)$, In "u_{eq}" Eqⁿ
R is $R_g = 287 \frac{J}{kg \cdot K}$. In other relations, $R = \frac{M_o}{M_u} \approx 2.5$ (in u(t))

Another hint: $\frac{\bar{D}}{M} = \frac{\int_0^t \frac{D}{M} dt}{t}$ & $\overline{g \cos \theta} = \frac{\int_0^t g \cos \theta dt}{t}$

in u(t) Eqⁿ, $u(t) = \dots - \frac{\int_0^t g \cos \theta dt}{t} - \frac{\int_0^t \frac{D}{M} dt}{t}$

In the hint of HW#1 in PDF, integral $g \cos \theta$ not $g \cos \theta \times \text{time}$
trapz(time, $g \cos \theta$) not trapz(time, $g \cos \theta \times \text{time}$)

Today lecture: Example (cont.), multi-stage rocket optimization

In the previous session, we considered a multi-stage rocket & found

$\lambda_i, \epsilon_i, \Delta u_{b, imp, t_i}$

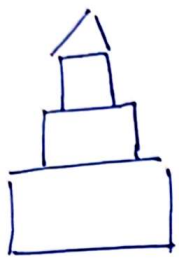
Now, consider a single-stage rocket with the same payload, same structural, same propellant masses as the previous 3-stage rocket: $M_o = 2,500,000 \text{ kg}$

$$M_s = M_{s_1} + M_{s_2} + M_{s_3} = 96,000 + 28,000 + 21,000 = 145,000 \text{ kg}$$

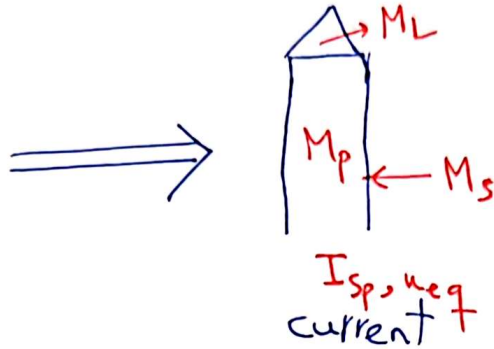
$$M_p = (1,824,000 + 373,000 + 90,000) = M_{p_1} + M_{p_2} + M_{p_3} = 2,287,000 \text{ kg}$$

(P 11)

$$M_L = 68,000 \text{ kg}, \quad I_{sp} = 370 \text{ s}, \quad u_{eq} = 3629.7 \frac{\text{m}}{\text{s}}$$



Previous



We converted the previous 3-stages rocket to a single-stage rocket with the same data!

Find $\lambda, \epsilon, \Delta u_{b, \text{imp}}$ for single-stage rocket.

Solution: $\lambda = \frac{M_L}{M_0 - M_L} = 0.92796$

$$\epsilon = \frac{M_S}{M_0 - M_L} = 0.05962$$

$$\Delta u_{b, \text{imp}} = u_{eq} \ln \frac{1+\lambda}{\epsilon+\lambda} = 8939 \frac{\text{m}}{\text{s}} < 11023 \frac{\text{m}}{\text{s}} \text{ "for 3-stages rocket"}$$

So $\Delta u_{b, \text{imp}}$ dropped! How much?

$$\% \Delta u_{b, \text{imp, improvement}} = \left\{ \frac{11023 - 8939}{8939} \times 100 \right\} = 23.3\%$$

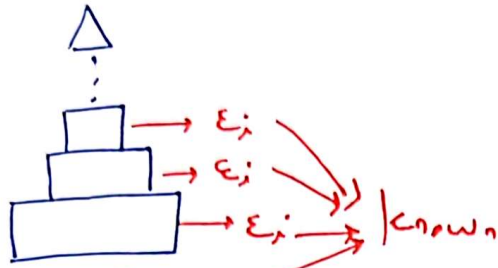
gain of staging!

"Multi-stage rocket optimization"

We want to maximize $\frac{M_L}{M_0}$ for a multi-stage rocket with the constraint of $\Delta u_{b, \text{imp}} = \text{fixed}$, number of stages, I_{sp} , & ϵ for each stage are known (fixed).

(P-2)

We want to adjust λ_i such that $(\frac{M_L}{M_o})$ becomes maximum & $\Delta u_{b, \text{imp}}$ etc.



$\Delta u_{b, \text{imp}} = \text{fixed}$

I_{sp_i}

We assumed $\Delta u_{b, \text{imp}} = \text{cte} \Rightarrow \delta [\Delta u_{b, \text{imp}}] = 0$
 Variation

$$\delta \left[\sum_{i=1}^n g_e I_{sp_i} \ln \left(\frac{1 + \lambda_i}{\epsilon_i + \lambda_i} \right) \right] = 0 \rightarrow \text{we can interchange } \delta \text{ \& } \Sigma$$

$$\Rightarrow \sum_{i=1}^n \delta \left[\underbrace{g_e I_{sp_i}}_{\text{known}} \ln \left(\frac{1 + \lambda_i}{\underbrace{\epsilon_i + \lambda_i}_{\text{known}}} \right) \right] = 0$$

unknown

From math:

$$d = \frac{\partial}{\partial x} dx + \frac{\partial}{\partial y} dy \quad (x-y \text{ domain}) \quad \text{so } \delta = \frac{\partial}{\partial I_{sp_i}} \delta I_{sp_i} + \frac{\partial}{\partial \epsilon_i} \delta \epsilon_i +$$

fixed & known!

$$\frac{\partial}{\partial \lambda_i} \delta \lambda_i \rightarrow \delta = \frac{\partial}{\partial \lambda_i} \delta \lambda_i$$

$$\Rightarrow \sum_{i=1}^n \frac{\partial}{\partial \lambda_i} \left[g_e I_{sp_i} \ln \left(\frac{1 + \lambda_i}{\epsilon_i + \lambda_i} \right) \right] \delta \lambda_i = 0$$

compute the derivative

$$\Rightarrow \sum_{i=1}^n \left[g_e I_{sp_i} \frac{(\epsilon_i - 1)}{(1 + \lambda_i)(\epsilon_i + \lambda_i)} \right] \delta \lambda_i = 0 \quad (A)$$

we should maximize $\frac{M_L}{M_0}$, so:

$$\frac{M_L}{M_0} = \prod_{i=1}^n \left(\frac{\lambda_i}{1+\lambda_i} \right), \text{ to maximize it, } \delta \left(\frac{M_L}{M_0} \right) = 0$$

$$\delta \left[\prod_{i=1}^n \left(\frac{\lambda_i}{1+\lambda_i} \right) \right] = 0. \text{ this is equivalent to:}$$

$$\delta \left[\sum_{i=1}^n L_n \left(\frac{\lambda_i}{1+\lambda_i} \right) \right] = 0 \implies \sum_{i=1}^n \left[\delta L_n \left(\frac{\lambda_i}{1+\lambda_i} \right) \right] = 0$$

$$\delta = \frac{\partial}{\partial \lambda_i} \delta \lambda_i \implies \sum_{i=1}^n \underbrace{\frac{\partial}{\partial \lambda_i} \left[L_n \left(\frac{\lambda_i}{1+\lambda_i} \right) \right]}_{\text{compute derivative}} \delta \lambda_i = 0$$

$$\sum_{i=1}^n \frac{1}{\lambda_i(1+\lambda_i)} \delta \lambda_i = 0 \quad \textcircled{B}$$

we have two relations, $\textcircled{A}$ from $\delta[\Delta u_{b, \text{simp}}] = 0$ & $\textcircled{B}$ from $\delta \left(\frac{M_L}{M_0} \right) = 0$.
 we want to maximize $\frac{M_L}{M_0}$ with the constraint $\textcircled{A}$ satisfied, so we introduce multiplier α :

$$\underbrace{\textcircled{B}}_{\text{target function}} + \alpha \underbrace{\textcircled{A}}_{\text{constraint}} = 0 \implies \text{linear combination of } \textcircled{B} \text{ \& } \textcircled{A} \text{ should be zero!}$$

$$\implies \sum_{i=1}^n \left[\frac{1}{\lambda_i(1+\lambda_i)} + \alpha g_{p_i} \frac{(E_i - 1)}{(1+\lambda_i)(E_i + \lambda_i)} \right] \delta \lambda_i = 0$$

(P-4)